Hands-on Exercise: Simple Python RAG

Duration: 90-120 minutes | Tool: PyCharm | Stack: Python, ChromaDB, local embeddings, AI API

Goal
Build a small RAG application that reads the two provided documents, stores their chunks in ChromaDB, retrieves matching text and answers questions using AI.

Task 1 - Read and tokenize

Create a PyCharm project and a **Documents** folder. Copy both supplied .txt files into it. Read and display the documents using Python. Use **tiktoken** to show the token count for each file.

Task 2 - Chunk and embed

Split both documents into chunks of **120 tokens** with **20-token overlap**. Print the chunk count and one sample chunk. Generate embeddings with **all-MiniLM-L6-v2** and display the vector size.

Task 3 - Store and search in ChromaDB

Create a persistent ChromaDB database inside the project. Save every chunk, its embedding and its source filename. Ask a question and display the **top 2 chunks** with their cosine distances.

Task 4 - Get an AI answer

Send the question and retrieved chunks to an AI model. Ask it to answer **only from the provided context**, or say it cannot find the answer. Print the final response.

Questions to test

- How many annual leave days do employees receive?
- Which days must employees work from the office?
- How quickly should a lost company laptop be reported?
- How quickly are Priority 1 incidents acknowledged?
- What is the company travel allowance per day?

Submission

Submit the Python files and screenshots showing: **chunks, ChromaDB search results, and 2 AI answers**. Keep the ChromaDB folder inside the project. **No web UI, FastAPI or Docker is required.**